

Consumer Demand

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The consumers Utility Maximization Problem (UMP)

Definition (Utility Maximization Problem)

Suppose that preferences over bundles $(x, y) \in \mathbb{R}_+^2$ are represented by $u(x, y)$, prices are (p_x, p_y) and income is I

$$\begin{array}{ll} \text{maximize} & u(x, y) \\ \text{subject to} & p_x x + p_y y \leq I \end{array} \quad (\text{UMP})$$

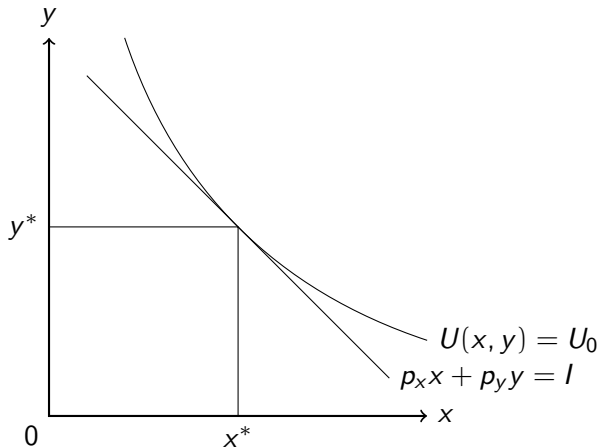
Graphical Analysis: Intuition

At an **interior** optimum, the slopes of the utility function and the budget line need to be equal. Thus

$$-\frac{p_x}{p_y} = -\frac{u_x}{u_y}$$

$$\frac{p_x}{p_y} = \frac{u_x}{u_y}$$

$$\frac{p_x}{p_y} = MRS$$



A detour on Unconstrained Optimization

Let f be a real-valued function defined over $\mathbb{R}^n$, that means $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition

The function f has a **local maximum** at x^* if there exists a neighborhood I such that it contains x^* and $f(x^*) \geq f(x)$ for all $x \in I$.

Definition

The function f has a **global maximum** at x^* if $f(x^*) \geq f(x)$ for all x in the domain of f .

First and Second order conditions: Single variable case

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued single variable function.

Theorem (First Order Necessary Condition for maximum/minimum)

Let f be a real-valued function of one variable. Suppose that f has a local maximum/minimum at some point x^ and suppose that $f'(x)$ exists. Then $f'(x^*) = 0$.*

Theorem (Second Order Sufficient condition for a maximum(minimum))

Let f be a real-valued function of one variable. Suppose that $f''(x^)$ exists and suppose that $f'(x^*) = 0$ and $f''(x^*) < 0$ ($f''(x^*) > 0$). Then f has a local maximum(minimum) at x^* .*

Multivariate function Unconstrained maximization

Theorem (First Order Necessary Condition for maximum/minimum)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Suppose that f has a local maximum/minimum at some point x^* and suppose that f is differentiable. Then $\frac{\partial f(x_1, \dots, x_n)}{\partial x_i} = 0$ for all $i = 1, \dots, n$

- To get second order sufficient conditions for maxima (minima), we need to be able to talk about when a multivariate function f is concave (convex).
- We will derive conditions on second order partial derivatives for this to happen. But first...

Theorem (Young's Theorem)

Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued multivariate function that has *continuous second order partial derivatives*. Then $f_{ij} = f_{ji}$ for $i \neq j$, $i, j = 1, 2, \dots, n$.

Second order conditions

- We will consider the case of a real-valued function of two variables $f(x, y)$.
- What conditions must we put on the partial derivatives to ensure that the change in f is negative when we move away from the critical point?
- The total differential is

$$df = f_x dx + f_y dy \quad (1)$$

- The total differential of (1) is

$$\begin{aligned} d^2f &= (f_{xx} dx + f_{xy} dy) dx + (f_{yx} dx + f_{yy} dy) dy \\ &= f_{xx} dx^2 + 2f_{xy} dx dy + f_{yy} dy^2 \quad \text{By Young's Theorem} \end{aligned} \quad (2)$$

- We want for d^2f to be unambiguously negative.
 - Suppose $dy = 0$, $d^2f < 0 \Rightarrow f_{xx} < 0$
 - Suppose $dx = 0$, $d^2f < 0 \Rightarrow f_{yy} < 0$
 - Suppose neither dx or dy are zero. Then we need $f_{xx}f_{yy} - f_{xy}^2 > 0$.

Back to constrained Optimization

Definition (First order conditions for interior maximum)

Suppose preferences are represented by the utility function $u(x, y)$. If (x^*, y^*) is an interior constrained maximum, then we must have

$$\frac{p_x}{p_y} = MRS(x^*, y^*)$$

Definition (Second order conditions for interior maximum)

A decreasing MRS is sufficient to guarantee that the point identified by the FOC $MRS = p_x/p_y$ is indeed the constrained maximum. What conditions on the utility function guarantee a decreasing MRS? If preferences are convex we have seen that they can be represented with a utility function which will have a decreasing MRS. What we are looking for is on conditions for the derivatives of u .

Optimization by Substitution

- Suppose that the consumer has **monotonic preferences**, i.e. the more the better.
- Then we know for sure that he will try to exhaust his income.
- Thus, the restriction of the budget constraint will hold with equality:
 $p_x x + p_y y = I$.
- Thus we can substitute $y = \frac{I - p_x x}{p_y}$ into the utility function to get

$$u(x, y) = u\left(x, \frac{I - p_x x}{p_y}\right)$$

- We can treat this problem as a single variable unconstrained maximization problem:

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy(x)}{dx} &= 0 \\ \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \frac{p_x}{p_y} &= 0 \\ \Rightarrow \frac{p_x}{p_y} &= \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = MRS\end{aligned}$$

- What about second order conditions?

Second Order Conditions

- Remember that we translated the problem from a multivariate **constrained** problem to a univariate **unconstrained** problem.
- FOC: $u_x(x, y(x)) - u_y(x, y(x)) \frac{p_x}{p_y} = 0$
- This condition is necessary, to get the sufficient condition, we need to examine changes away from this critical point.
- We need to obtain the second derivative and make sure it is negative.

$$\left[u_{xx} - u_{xy} \frac{p_x}{p_y} \right] - \frac{p_x}{p_y} \left[u_{yx} - u_{yy} \frac{p_x}{p_y} \right] < 0$$

but at the candidate optimum, $\frac{p_x}{p_y} = \frac{u_x}{u_y}$, and $u_{xy} = u_{yx}$ by Young's Thm so

$$\begin{aligned} \left[u_{xx} - u_{xy} \frac{u_x}{u_y} \right] - \frac{u_x}{u_y} \left[u_{yx} - u_{yy} \frac{u_x}{u_y} \right] &< 0 \\ \frac{u_{xx} u_y^2 - 2u_{xy} u_x u_y + u_x^2 u_{yy}}{u_y^2} &< 0 \end{aligned}$$

Which implies that: $u_{xx} u_y^2 - 2u_{xy} u_x u_y + u_x^2 u_{yy} < 0$

First Order Necessary and Sufficient condition for an interior optimum

- The assumption of the more the better means that Marginal utilities are strictly positive, i.e. $u_x > 0$ and $u_y > 0$. Intuitively, the more I consume of a good, the more “utility” I get.
- $u_{xx}u_y^2 - 2u_{xy}u_xu_y + u_x^2u_{yy} < 0$ is the formal condition that guarantees that the MRS is decreasing. We have seen previously that this is equivalent to saying that the preference relation is convex.
- $u_{xx}u_y^2 - 2u_{xy}u_xu_y + u_x^2u_{yy} < 0$ is the formal condition that guarantees “The consumer strictly prefers more balanced bundles than extreme ones.”

Theorem

Suppose u is a differentiable utility function that represents monotone and convex preferences, i.e. the conditions

- $u_x > 0$ and $u_y > 0$,
- $u_{xx}u_y^2 - 2u_{xy}u_xu_y + u_x^2u_{yy} < 0$ hold. Then, the *necessary* and sufficient condition for the utility maximization problem is:

$$\frac{p_x}{p_y} = \frac{u_x}{u_y}$$

Example: Cobb-Douglas Utility function

Cobb-Douglas utility function

Consider the problem of maximizing $u(x, y) = x^\alpha y^{1-\alpha}$, where $\alpha > 0$

$$\begin{array}{ll} \text{maximize} & x^\alpha y^{1-\alpha} \\ \text{subject to} & p_x x + p_y y \leq I \end{array} \quad (\text{Cobb-Douglas UMP})$$

- Consider the monotonic transformation $f(u) = \ln(u)$
- Marginal utilities are $u_x = \alpha/x$, $u_y = (1 - \alpha)/y$ which are both positive.
- $u_{xx} = -\alpha/x^2$, $u_{yy} = -(1 - \alpha)/y^2$, $u_{xy} = 0$ so, indifference curves are convex.
- Then the first order condition identifies an optimum

$$\frac{p_x}{p_y} = \frac{\alpha}{1 - \alpha} \frac{y}{x}$$

Cobb-Douglas example, continued

- Solving for y in the first order condition yields:

$$y = \frac{p_x}{p_y} \frac{1 - \alpha}{\alpha} x$$

- Plugging in the budget line and solving for x yields:

$$x^M(p_x, p_y, I) = \frac{\alpha I}{p_x}$$

- Similarly, we can get

$$y^M(p_x, p_y, I) = \frac{(1 - \alpha)I}{p_y}$$

- x^M and y^M are called the Marshallian or uncompensated demands.

What can go wrong? Corner solutions, Bliss points

- The condition $MRS = p_x/p_y$ helps identify an interior optimum, under the assumption of differentiability.
- Non-differentiable preferences, Perfect Complements.
- What happens when it is optimal to consume positive quantities of one good, but none of the other?
- Examples, perfect substitutes.
- On the other hand, what if the preference do not satisfy the more the better?
- If preferences have a **bliss point** it is possible that the optimum can happen **inside the budget set**.
- We need to amend the first order condition to deal with these cases.